

#1 Radiation transport

#1a: 8 points total

$$(i) K_{es} = 0.4 \text{ cm}^2/\text{g} ; \bar{\rho}_0 = \frac{M_0}{V_0} = \frac{3}{4\pi} \frac{M_0}{R_0^3}$$

$$l_{\text{photon},0} = \frac{1}{K_{es} \bar{\rho}_0} = \left[\frac{1}{0.4 \frac{\text{cm}^2}{\text{g}}} \right] \left[\frac{1}{1.419/\text{cm}^3} \right]$$

$$\boxed{l_{\text{photon},0} = 1.77 \text{ cm}}$$

1 pt attempt ; 1 pt correct #

$$(ii) v_{\text{photon}} = C = [\text{length} / \text{time}] = 3 \times 10^{10} \frac{\text{cm}}{\text{s}}$$

$$l_{\text{photon},0} = 1.77 \text{ cm} = [\text{length}]$$

$$R_0 = 6.96 \times 10^{10} \text{ cm} = [\text{length}]$$

of steps required to travel distance R_0 is

$$N = \left(\frac{R_0}{l_{\text{photon},0}} \right)^2$$

$$\text{Total path traveled is } l_{\text{photon},0} \cdot N = \frac{R_0^2}{l_{\text{photon},0}} = [\text{length}]$$

The time it takes to travel that path is

$$t = \frac{l_{\text{photon},0} N}{C} = \frac{R_0^2}{l_{\text{photon},0} C} = 9 \times 10^{10} \text{ s} \approx 300 \text{ yr}$$

+ 3 pts for attempt ; + 1 pt for correct

$$(iii) \tau_{\kappa,0} = \frac{GM_0^2}{2R_0 L_0} \approx 1.5 \times 10^7 \text{ yr}$$

$$\rightarrow \frac{\tau_{\text{photon},0}}{\tau_{\kappa,0}} = \frac{3000}{1.5 \times 10^7} = 2 \times 10^{-4}$$

much shorter !

+1 for attempt ; +1 for correct

#1b : 2 pts for ≥ 1 sentence; 2pts for correct interp.

The diffusion approximation requires that the mean free path be much smaller than the size of the star. However, near the surface, the density decreases significantly so the mean free path increases, until eventually photons begin escaping the surface.

$$\begin{aligned} \#1c : \frac{\partial T}{\partial r} &= -\frac{3\kappa\rho}{16\pi ac T^3} \frac{l}{r^2} ; T \approx 10^7 \text{ K}, r \approx 0.2 R_0, l \approx 3.8 \times 10^{33} \text{ erg/s} \\ &\quad \kappa = 0.4 \text{ cm}^2/\text{g} ; \bar{\rho} = 1.4 \text{ g/cm}^3 \\ &= -\frac{3 [0.4 \text{ cm}^2/\text{g}] [1.4 \text{ g/cm}^3] [3.8 \times 10^{33} \text{ erg/s}]}{16\pi [7.56 \times 10^{-15} \text{ erg/cm}^3 \text{K}^4] [3 \times 10^{10} \text{ cm/s}] [10^7 \text{ K}]^3 [6.96 \times 10^{10} \text{ cm}^2 \cdot 0.2]^2} \end{aligned}$$

$$\boxed{\frac{\partial T}{\partial r} = 3 \times 10^{-6} \text{ K/cm}}$$

#1c continued

$$\Delta T = \frac{\partial T}{\partial r} \ell_{\text{photon}} = \left(3 \times 10^{-6} \frac{\text{K}}{\text{cm}} \right) (1.77 \text{ cm})$$
$$= \boxed{5 \times 10^{-6} \text{ K} = \Delta T}$$

This is a tiny temperature gradient!

+2 for $\frac{\partial T}{\partial r}$ attempt; +1 for correct

+2 for ΔT attempt; +1 for correct

#2: 20 points; +10 for figure; +10 for caption.

-1 for each incorrect/missing axis labels

-1 for each missing axis/data description

-1 for missing punchline.